



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO VIJAY TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

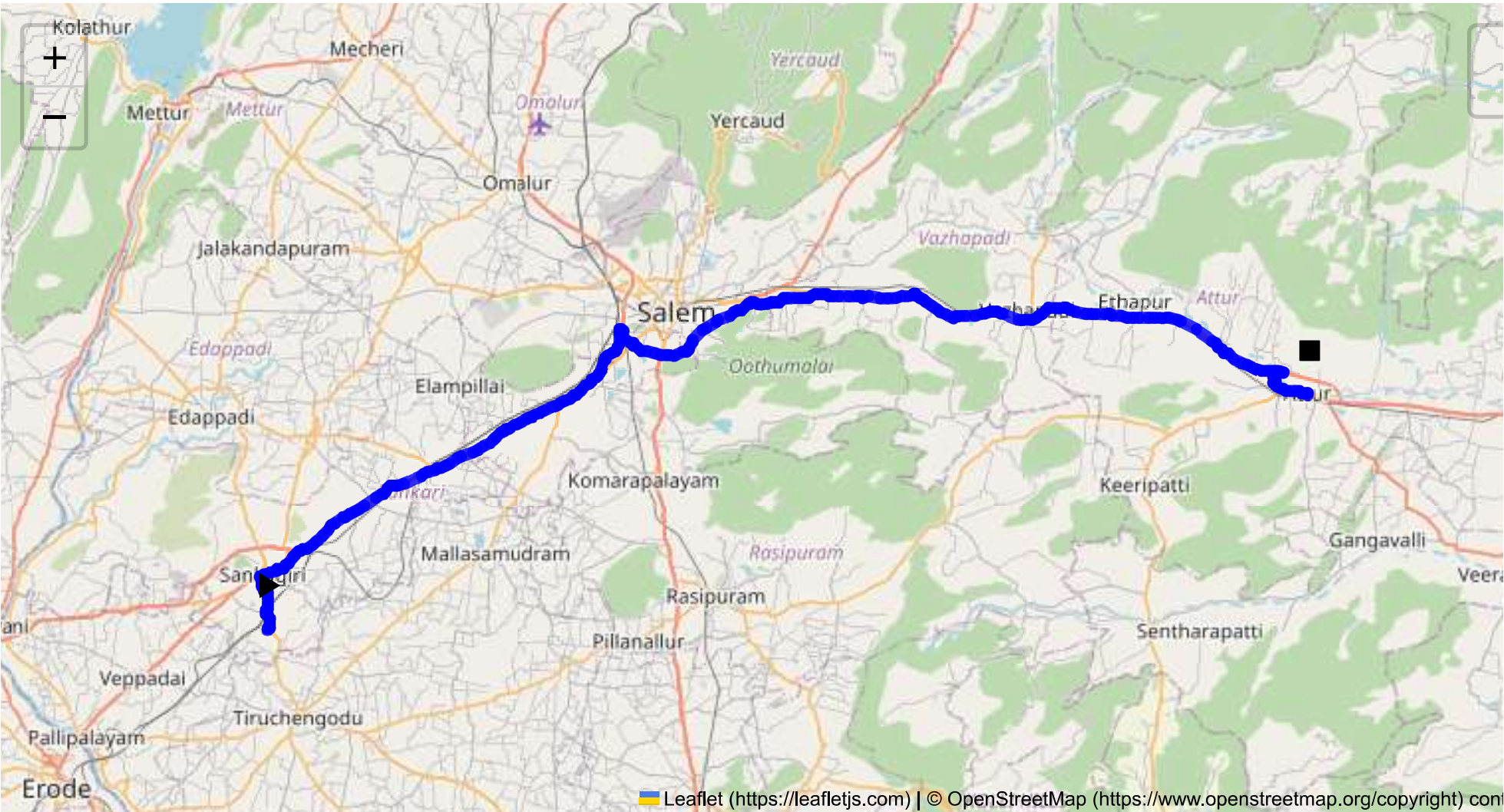
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 98.34 km
Estimated Duration: 1.9 hours
Adjusted Duration (Heavy Vehicle): 2.4 hours
Start: (11.4381, 77.8734)
End: (11.59868, 78.596336)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Attur via CVQF+23W Sangagiri and Udayarpalayam covers approximately 98.34 kilometers, traversing mainly through state highways and local roads. The journey involves driving through rural landscapes dotted with small towns and agricultural fields, ensuring a mix of higher-speed sections and areas requiring careful navigation through populated regions.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Conditions:** Tamil Nadu primarily experiences tropical weather conditions. The route can be affected by monsoon rains from June to September, causing potential flooding and slippery roads.
- **Hazards:** Heavy rainfall may lead to poor visibility and water-logged roads. During summer months, temperatures can soar, potentially affecting vehicle performance.

3. Analysis of Traffic Patterns

- **Peak Hours:** Traffic typically peaks in the mornings (8 AM to 10 AM) and evenings (6 PM to 8 PM).
- **Congestion-Prone Areas:** Expect congestion near towns like Sankari and Edappadi, particularly around market areas and intersections.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** The state highways are generally well maintained; however, roads may narrow near rural sections.
- **Infrastructure:** Adequate infrastructure with petrol stations and rest areas along the highway. Some stretches may lack adequate lighting, especially in less developed regions.

5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route:** For emergencies, consider taking the NH79, which parallels parts of this route. Although slightly longer, it offers quicker access to emergency services.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Regulations:** Tamil Nadu mandates special permits for transporting hazardous materials, including following designated routes and times.
- **Restrictions:** Night curfews in certain towns may affect travel plans. Ensure compliance with signage and restrictions.

7. Overview of Historical Incidents

- This region has a moderately low incidence rate of heavy vehicle accidents; however, incidents have been reported primarily due to over-loading and brake failures on inclines.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** Be mindful around agricultural zones to avoid environmental pollution. Limited wildlife crossings may exist, especially near forested patches.
- **Pollution Controls:** Adhere to strict emission norms and avoid idling in residential areas.

9. Analysis of Communication Coverage

- **Coverage:** Most of the route has good mobile network coverage, though intermittent dead zones may occur in isolated rural areas.
- **Dead Zones:** Likely near heavily forested regions and valleys.

10. Estimated Emergency Response Times

- **Response Times:** Along main highways, expect emergency response times of approximately 20-30 minutes; this may extend to 45 minutes in remote sections.

11. Overall Summary of Risk Assessment

- **Risk Level:** Moderate. The route is mostly secure but requires attention to weather conditions, rural traffic patterns, and adherence to local transport regulations.
- **Recommendations:** Proper planning with allowances for emergency detours and understanding peak congestion areas can mitigate most risks. Maintain communication devices with fully charged batteries, and utilize GPS for real-time updates to navigate traffic and environmental conditions.

Overall, while the route presents typical challenges encountered in rural travels, adherence to safety protocols and preparedness through information and regulation compliance will ensure a safer transit of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44900, 77.87406	30 KM/Hr
4	Turn	Medium	11.45000, 77.87213	30 KM/Hr
5	Turn	High	11.45028, 77.87202	15 KM/Hr
6	Turn	Medium	11.45309, 77.87325	30 KM/Hr
7	Turn	Medium	11.46400, 77.87323	30 KM/Hr
8	Turn	Medium	11.46765, 77.87037	30 KM/Hr
9	Turn	High	11.47403, 77.86841	15 KM/Hr
10	Turn	High	11.49149, 77.89601	15 KM/Hr
11	Turn	High	11.49224, 77.89601	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	Medium	11.64172, 78.11817	30 KM/Hr
13	Turn	Medium	11.64216, 78.11875	30 KM/Hr
14	Turn	Medium	11.64195, 78.12129	30 KM/Hr
15	Turn	Medium	11.64162, 78.12160	30 KM/Hr
16	Turn	Medium	11.64120, 78.12168	30 KM/Hr
17	Turn	Medium	11.63948, 78.12081	30 KM/Hr
18	Turn	Medium	11.65441, 78.38090	30 KM/Hr
19	Blind Spot	Blind Spot	11.61369, 78.57920	10 KM/Hr
20	Turn	High	11.61426, 78.57274	15 KM/Hr
21	Turn	Medium	11.61260, 78.57264	30 KM/Hr
22	Turn	Medium	11.61237, 78.57287	30 KM/Hr
23	Turn	Medium	11.60852, 78.57382	30 KM/Hr
24	Turn	Medium	11.60476, 78.57394	30 KM/Hr
25	Turn	Medium	11.60442, 78.57618	30 KM/Hr
26	Turn	High	11.59965, 78.59643	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
1	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
13	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
14	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
17	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
21	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
23	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
24	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
25	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
26	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
27	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
28	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
29	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
30	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
31	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
32	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
36	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
38	clinic	Lavanya Clinic	11.652121, 78.4795146	30 km/h	Medium
39	hospital	Amsh Agathiyar Sidha Hospital	11.650564, 78.483542	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
40	clinic	Sudha Polyclinic	11.647825, 78.508537	30 km/h	Medium
41	hospital	Government Hospital, Kothampadi	11.62641, 78.542199	30 km/h	Medium
45	hospital	Sundaram Multi Speciality Hospital	11.600633, 78.5948143	30 km/h	Medium
46	clinic	RK Clinic	11.59969, 78.5941386	30 km/h	Medium
47	clinic	Mother's Ortho Clinic	11.6002438, 78.594053	30 km/h	Medium
48	hospital	Chiranjeevi Hospital	11.5998283, 78.594309	30 km/h	Medium
49	clinic	Sri Vinayaga Physiotherapy Clinic	11.598134, 78.597401	30 km/h	Medium
50	hospital	Sri Andavhar Nursing Home	11.6010351, 78.5974614	30 km/h	Medium
51	clinic	Attur ENT and Dental Care Centre	11.5991073, 78.5962503	30 km/h	Medium
52	hospital	Kalyani Memorial Hospital	11.599983, 78.596223	30 km/h	Medium
53	clinic	SS Clinic	11.5975084, 78.5979036	30 km/h	Medium
54	hospital	Gokulam Hospital	11.597356, 78.598403	30 km/h	Medium
55	hospital	Arun Hospital	11.599848, 78.596343	30 km/h	Medium
56	clinic	Bawaa Clinic	11.597934, 78.597143	30 km/h	Medium
57	hospital	Kumar Eye Hospital	11.5978065, 78.5984006	30 km/h	Medium
58	clinic	Ramakrishna Nursing Home	11.5988349, 78.5988615	30 km/h	Medium
59	hospital	Anugraha Hospital	11.5963566, 78.5941161	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
15	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
16	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
18	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
19	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
20	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
22	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
33	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
34	school	Thangam ZEE public school	11.6653353, 78.251641	30 km/h	Medium
35	college	AVS College od Arts and Science, Salem.	11.6642672, 78.2584134	30 km/h	Medium
37	college	Mahendra Engineering College	11.6672537, 78.2641303	30 km/h	Medium
42	school	Vani Vikas Matric Hr.sec school	11.6212713, 78.5462658	30 km/h	Medium
43	school	Government Boys Higher Secondary School	11.598743, 78.5885472	30 km/h	Medium
44	school	Government Boys Higher Secondary School	11.599149, 78.5888931	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

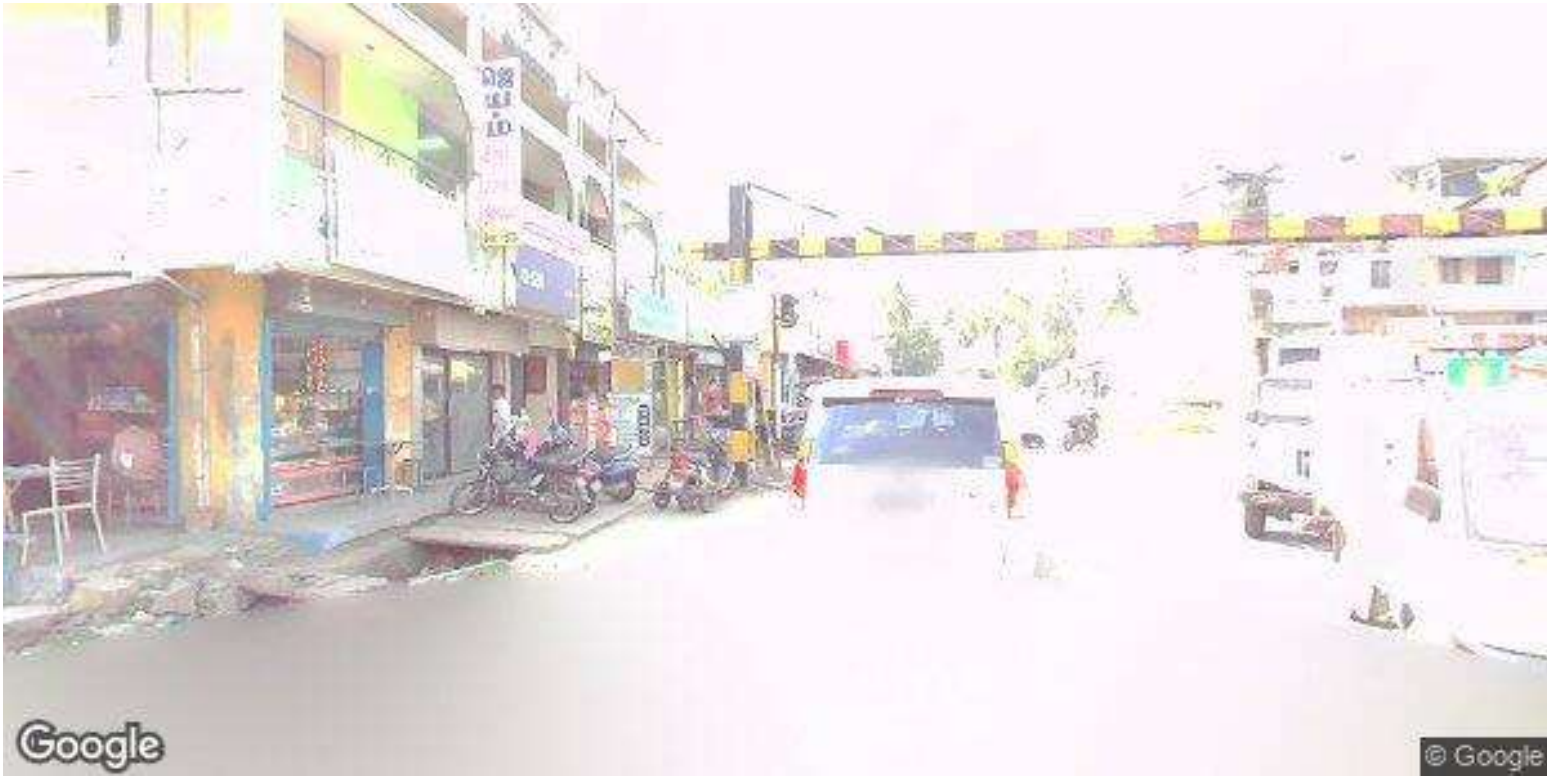


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45000, 77.87213



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45309, 77.87325



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64172, 78.11817



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64216, 78.11875



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64195, 78.12129



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64162, 78.12160



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64120, 78.12168



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63948, 78.12081



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65441, 78.38090



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61369, 78.57920



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.61426, 78.57274



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61260, 78.57264



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61237, 78.57287



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60852, 78.57382



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60476, 78.57394



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60442, 78.57618



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59965, 78.59643